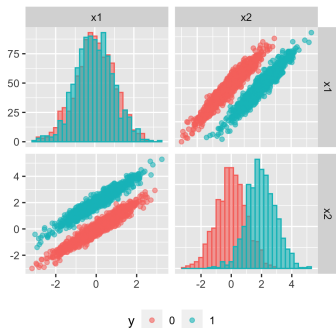


Introduction to Machine Learning

Feature Selection

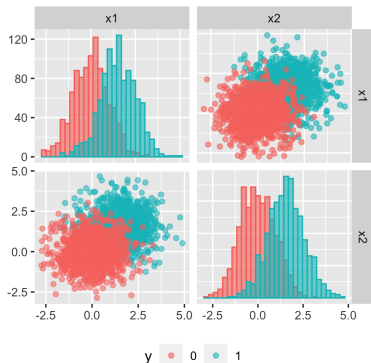
Feature Selection: Filter Methods (Examples and Caveats)



Learning goals

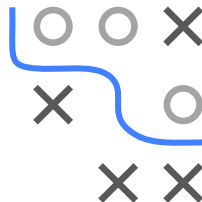
- Understand how filter methods can be misleading
- Understand how filters can be applied and tuned

FILTER METHODS CAN BE MISLEADING



ρ_{ACC} of log. reg. classifier with:

- feature x_1 : 0.76
- feature x_2 : 0.78
- both features: 0.85



Information gain from presumably redundant variables. 2 class problem with indep features. Each class has Gaussian distribution with no covariance. While filter methods suggest redundancy, combination of both vars yields improvement, showing indep vars are not truly redundant. For further details, see [► Guyon and Elisseeff 2003](#).

A 3x3 grid with a blue path starting at the top-left cell (0,0) and ending at the bottom-right cell (2,2). The path consists of the following cells: (0,0), (0,1), (1,1), (1,2), and (2,2). The cells (0,2), (1,0), and (2,0) are empty. The cells (0,1), (1,1), and (1,2) contain a grey 'X'. The cells (1,0) and (2,0) contain a grey circle.

-
- Performance of classif.kknn on dataset har
- Y-axis: classif.mbrrier (ranging from -0.05 to 0.60)
- X-axis: Number of Features (ranging from 0 to 500)
- Legend: ● F-test (red), ● information gain (cyan)
- | Method | Number of Features | classif.mbrrier |
|------------------|--------------------|-----------------|
| F-test | 210 | 0.041 |
| information gain | 288 | 0.042 |

Feature Score and log_reg model performance. Dataset: SPAM

-
- Figure 1 is a line plot showing the relationship between the number of features used in a model and two performance metrics: Information Gain and AUC. The x-axis represents the 'Feature rank/Features used in the model' from 0 to 40. The left y-axis represents 'Information Gain' from 0.00 to 0.03. The right y-axis represents 'AUC' from 0.86 to 0.94. The blue line with dots represents Information Gain, which starts at approximately 0.035 for 1 feature and decreases rapidly, reaching near zero by feature rank 20. The red line with dots represents AUC, which starts at approximately 0.85 for 1 feature and increases, peaking around 0.95 at feature rank 40.